



Treewidth and logical definability of graph products

Selma Djelloul

LRI, UMR 8623, Bât 490 Université de Paris-Sud, 91405 Orsay Cedex, France

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ABSTRACT

In this paper we describe an algorithm that, given a tree-decomposition of a graph G and a tree-decomposition of a graph H , provides a tree-decomposition of the cartesian product of G and H . Using this algorithm, we derive upper bounds on the treewidth (resp. on the pathwidth) of the cartesian product of two graphs, expressed in terms of the treewidth (resp. pathwidth) and the size of the factor graphs. In the context of graph grammars and graph logic, we prove that the cartesian product of a class of graphs by a finite set of graphs preserves the property of being a context-free set, and that the cartesian product by a finite set of connected graphs preserves MS_1 -definability and MS_2 -definability. We also prove that the cartesian product of two MS_2 -definable classes of connected graphs is MS_2 -definable.

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1. Introduction

In this paper, we consider cartesian products of graphs. All graphs considered here are finite and have no loops. Given two graphs G and H , their cartesian product, $G \otimes H$, is the graph whose vertex set is the cartesian product $V_G \times V_H$. For every pair u, u' of adjacent vertices of G , for every vertex $v \in V_H$, vertices (u, v) and (u', v) are connected in $G \otimes H$ using as many edges as the multiplicity of the edge connecting u and u' in G . For every pair v, v' of adjacent vertices of H , for every vertex $u \in V_G$, vertices (u, v) and (u, v') are connected in $G \otimes H$ using as many edges as the multiplicity of the edge connecting v and v' in H . Grids and hypercubes are among the most popular families of cartesian product of graphs. The interested reader is referred to the book [11] that is exclusively dedicated to results on graph products. Our research is motivated by the following:

- questions about parameters of graphs related to structural decompositions: treewidth, pathwidth, etc;
- questions about grammatical descriptions of graphs;
- questions related to *monadic second-order logic* ($MSOL$) and decidability of MS -theories of certain classes of graphs;

with application to classes of graphs obtained by natural operations such as product of graphs.

We now present in more detail the theoretical background of our contributions with its two main motivations: (1) polynomial algorithms, and (2) decidability of logical theories.

In the early eighties, Robertson and Seymour published their first results of their *Graph Minor* series of papers. They defined the concept of *tree-decomposition* and *treewidth*. This notion has come to play an important role in recent investigations in algorithmic graph theory. Tree-decomposition is equivalent to the notion of *partial k -tree* which has already been used, directly or indirectly, in many algorithmic constructions designed independently. Courcelle [5] established a unified framework for many graph properties that were considered separately by several authors. He proved that every graph property that can be expressed in *monadic second-order logic* ($MSOL$) is decidable in linear time (on the number of vertices) for simple graphs of bounded treewidth. The reference book [10] gives a framework to study the links between on

E-mail address: djelloul@lri.fr.

one hand results by Robertson and Seymour, Courcelle, and other authors, and on the other hand algorithmic *parameterized* complexity.

Determining whether a graph has a treewidth at most k is *NP*-complete if k is part of the input, but decidable in linear time for fixed k (Bodlaender [2]). In case of a positive answer, a tree-decomposition is even produced by the algorithm in [2]. However, this linear time algorithm is not practical because of the large size of constants depending on k .

Bauderon and Courcelle [1] defined operations that enable viewing the set of all finite (hyper)-graphs as a *many sorted algebra* over an infinite signature. Such algebras have been extensively studied, particularly those defined by HR-(*hyperedge replacement*) and VR-(*vertex replacement*) signatures. In this paper we are focussing on HR grammars. The key interest of HR grammars is that, by such algebraic operations, one can generate all finite graphs of treewidth at most k , for fixed k , from elementary graphs [6]. This can be done in a way that can be compared with context-free grammars defining words. HR-equational sets of graphs have been defined by means of systems of recursive equations (see, e.g., [9]). An equational set is defined as the least solution (for set-inclusion) of a component (i.e., an equation) of such a system. For the reader who is not familiar with these notions, we recall the notions of graph algebras and equational sets of graphs in the appendix. Graphs that are members of such sets have a derivation tree according to the grammar. This often enables one to design proofs by induction on the size of the derivation tree. HR-equational sets of graphs have another interest related to MSOL. This logic is first-order logic on power-sets. Hence, in case of a graph considered as a relational structure with the set of vertices as the domain, and using the binary relation of adjacency, quantified variables can denote sets of vertices. We use the same notations as the ones introduced by Courcelle, and denote by MS_1 this variant of MSOL. In the MS_2 variant, quantified variables can denote sets of edges because the domain also includes the set of edges. A set of graphs L has a decidable MS_i -theory iff there exists an algorithm that decides whether a given MS_i -formula is satisfied by every graph of L . Since each of the two logical languages is closed under negation, asking whether a formula is satisfied by all the graphs of the class is the same as asking whether a formula is satisfied by some graph in the class. Hence the decidability of MS_i -theories of a graph class is equivalent to the decidability of the emptiness problem of MSOL over this class. The MS_2 -theory of an HR-equational set of graphs is decidable [5]. A similar result relates decidability of MS_1 -theory and VR-equational sets.

One of the major incidence of Courcelle's developments on graph grammars and monadic second order logic for graphs is that, if k is fixed, the class of all graphs with treewidth at most k is HR-equational; hence, it has a decidable MS_2 -theory. A class of graphs having a decidable MS_2 -theory is of bounded *treewidth* [13]; the converse does not hold.

A set of graphs is MS_i -definable iff it is the set of graphs satisfying an MS_i -formula. The intersection of an HR-equational set and an MS_2 -definable set is HR-equational [5]. As a consequence, if a class of graphs with bounded treewidth is MS_2 -definable then its MS_2 -theory is decidable.

Our results

We prove a new upper bound on the treewidth of the cartesian product expressed in terms of the treewidths and the size of the two involved graphs. This bound is obtained by means of two algorithmic procedures. We obtain refinements of upper bounds on the treewidth of certain classes of cartesian products. We use Chlebíková's result [3] stating that a grid $G_{p \times q}$ has treewidth $\min\{p, q\}$ to derive a graph class of cartesian products containing the class of grids, and with the property that each of its elements has treewidth $\min\{p, q\}$ where p and q are the number of vertices of the two involved graphs. We also derive from our general construction an upper bound on the pathwidth of the cartesian product expressed in terms of the pathwidths of the two involved graphs.

Regular grammars on words generate languages that are recognized by finite state automata. There is no convenient notion of finite automaton for graphs. Nevertheless, it follows from results by Courcelle that a monadic second order formula is finite state for graphs of bounded treewidth. Hence an MS -definable class of graphs can be viewed in a way comparable with a regular language on words. Furthermore, from the aforementioned results, we deduce that an MS formula acts, for a fixed k , as a filter that extracts a subclass that has a decidable MS_2 -theory from the class of all graphs with *treewidth* at most k . Our contributions in this context, are:

- The cartesian product by a finite class of graphs preserves HR-equationality,
- The cartesian product by a finite class of connected graphs preserves MS_1 and MS_2 -definability.
- The cartesian product of two MS_2 -definable classes of connected graphs is MS_2 -definable.

2. Definitions and notation

Definition 2.1 (*Tree-Decomposition, Path-Decomposition*). Let $G = (V, E)$ be a graph. A *tree-decomposition* of G is a pair $(T, (X_t)_{t \in V_T})$ where T is a tree and $(X_t)_{t \in V_T}$ a family of subsets of V_G , with the following properties:

(P1) $\bigcup_{t \in V_T} X_t = V_G$.

(P2) For every edge e of G there exists $t \in V_T$ such that e has both ends in X_t .

(P3) For t, t', t'' , if t' is on the path between t and t'' then $X_t \cap X_{t''} \subseteq X_{t'}$.

The *width* of the tree-decomposition is $\max_{t \in V_T} (|X_t| - 1)$.

The graph G has *treewidth* w if w is the smallest integer such that G has a tree-decomposition of width w . We write $\text{twd}(G) = w$.

If T is a path, the decomposition is called a *path-decomposition*. The graph G has *pathwidth* w if w is the smallest integer such that G has a path-decomposition of width w . We write $\text{pwd}(G) = w$.

Every tree-decomposition can be transformed into a tree-decomposition of the same width in which all pairs $X_t, X_{t'}$, where t, t' are adjacent nodes, either strictly intersect or are disjoint. The transformation can be performed in linear time in the number of nodes of T by repeatedly contracting an edge $[t, t']$ of T such that $X_t \subseteq X_{t'}$ or $X_{t'} \subseteq X_t$ until this operation becomes impossible. A tree-decomposition with such a property is said to be *irredundant*.

A decomposition which is irredundant and of the minimum width is said to be *quasi-fundamental*. In the literature (e.g., [3]), a tree-decomposition is said to be *fundamental* if it satisfies the stronger condition of being of the minimum width and with the minimum number of nodes of T .

We use the following notation :

$\text{TWD}(=k)$ (resp. $\text{PWD}(=k)$) denotes the set of all graphs of treewidth (resp. pathwidth) k .

Definition 2.2 (*Parent and Descendent Bags Mappings*). Let $(\mathcal{B}, (Y_r))$ be a tree-decomposition of a graph G rooted at Y_{r_0} .

- i. We define the mapping *parent-bag* $p : V_{\mathcal{B}} \rightarrow \mathcal{P}(V_G)$ such that, for $r \neq r_0, p(r) = Y_{r'}$, where r' is the parent of r in the rooted tree $\mathcal{B}$, and $p(r_0) = \emptyset$.
Note that, if the decomposition is irredundant, there always exists $y \in Y_r - p(r)$.
- ii. We define the mapping *descendent-bags* $d : V_{\mathcal{B}} \rightarrow \mathcal{P}(V_G)$ such that for every internal node $r, d(r) = \bigcup_{\alpha, r_\alpha \text{ child of } r} Y_{r_\alpha}$, and $d(r) = \emptyset$ if r is a leaf.

Definition 2.3 (*Bag Types*). Let $(\mathcal{B}, (Y_r))$ be a rooted irredundant tree-decomposition of width w of a graph G .

Relax-introduce. We say that a bag Y_r is *relax-introduce* if $|Y_r| = w + 1$, and there exist two distinct vertices $y \in Y_r - p(r)$ and $z \in Y_r - d(r)$. Then, a bag Y_r is *relax-introduce* if $|Y_r|$ is tight and there exists a vertex z that comes into Y_r just when another vertex leaves Y_r . We say that the couple (y, z) is *synchronized* and that the vertex z is the *partner* of the leaving vertex y .

Furtive-introduce. We say that a bag Y_r is *furtive-introduce* if $|Y_r \cap p(r)| = w$ and $Y_r - p(r) \not\subseteq d(r)$. Note that, $|Y_r \cap p(r)| = w$ implies that $|Y_r - p(r)| = 1$. Let $\{y\} = Y_r - p(r)$. This vertex is called the *furtive* vertex. Then, a bag Y_r is *furtive-introduce* if $|Y_r \cap p(r)|$ is tight and the leaving vertex does not come from $d(r)$.

Single-branch-leaving. Let r be a node with at least two children. We say that Y_r is *r_β -single-branch-leaving* if r_β is a child of $r, |Y_r| = w + 1, Y_r \subseteq d(r)$, and there exists at least one vertex $y \in Y_r - p(r)$ for which r_β is the unique child such that $y \in Y_{r_\beta}$. In this case, such a vertex y is called the *single-branch-leaving* vertex. Then, a bag Y_r is *single-branch-leaving* if it has at least two children, $|Y_r|$ is tight, $Y_r \subseteq d(r)$, and at least one leaving vertex comes from a single branch of the descendance of Y_r . Since the decomposition is irredundant, there exists $x \in Y_r - Y_{r_\beta}$. We say that the couple (y, x) is *synchronized* and that the vertex x is the *partner* of the leaving vertex y .

Blocking. A bag Y_r is *blocking* if r has at least two children, $|Y_r| = w + 1, Y_r \subseteq d(r)$, and for every vertex $y \in Y_r - p(r)$, there are at least two distinct children r_β, r_γ of r such that both Y_{r_β} and Y_{r_γ} contain y .

Remark 2.4. Let $(\mathcal{B}, (Y_r))$ be a rooted irredundant tree-decomposition of width w of a graph G .

- i. A clique on n vertices, $n \geq 2$, has a unique rooted fundamental tree-decomposition reduced to a single bag that is *relax-introduce*.
- ii. A bag can be both *relax-introduce* and *furtive-introduce*.
- iii. Every bag of size $w + 1$ is of (at least) one of the types described above.
- iv. Types *relax-introduce* (resp. *furtive-introduce*), *single-branch-leaving* and *blocking* are mutually excluding.
- v. A leaf bag cannot be *blocking* and cannot be *single-branch-leaving*.
- vi. If $w \geq 1$, the root bag cannot be *furtive-introduce*.
- vii. If r is a node with a single child such that $|Y_r| = w + 1$, then Y_r is *relax-introduce* or *furtive-introduce* (possibly both). Therefore, every graph G such that $\text{pwd}(G) = \text{twd}(G)$, has at least one quasi-fundamental decomposition with no blocking bag. This includes grids and AT-free graphs (Asteroidal Triple), the largest known class of graphs for which the pathwidth equals the treewidth [12].

Definition 2.5 (*Compounds of a Tree by a Tree*). In the following $\mathcal{B}$ and $\mathcal{A}$ denote disjoint rooted trees, and r denotes a node of $\mathcal{B}$.

- A *compound* of $\mathcal{B}$ by $\mathcal{A}$ at r is the rooted tree obtained by replacing r by a copy of $\mathcal{A}$, and, if r is not the root, connecting the root of $\mathcal{A}$ to the parent of r . We also make each child of r (if any) a child of an arbitrary node of $\mathcal{A}$. We denote such a resulting tree by $\mathcal{B}[r \downarrow \mathcal{A}]$.
- A *compound* of $\mathcal{B}$ by $\mathcal{A}$ at r is said to be *compatible* if all the children of r (if any) are made the children of the same arbitrary leaf f of $\mathcal{A}$. We denote such a resulting tree by $\mathcal{B}[r \downarrow \mathcal{A}]_f$.

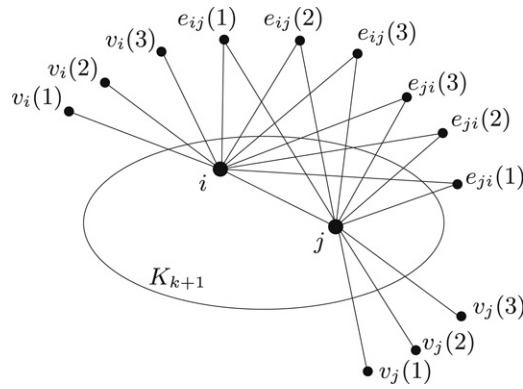


Fig. 1. The graph $G_{k,3}$. Only the neighborhood of the edge $[i, j]$ of K_{k+1} is represented.

- If r has at least two children, an r_β -single-branch compound of $\mathcal{B}$ by $\mathcal{A}$ at r , where r_β is a child of r , is a compound in which r_β is made the child of an arbitrary leaf of $\mathcal{A}$ and all the other children of r are made children of the root of $\mathcal{A}$. We denote such a resulting tree by $\mathcal{B}[r \downarrow_{r_\beta} \mathcal{A}]$.

For each type of compound, we rename each node s of $\mathcal{A}$ to (r, s) in the resulting tree where $r \in V_{\mathcal{B}}$ is the node at which the compound occurs.

3. Treewidth and pathwidth for cartesian product of graphs

If $(T, (X_t)_{t \in V_T})$ is a tree-decomposition of G then $(T, (X_t \times V_H)_{t \in V_T})$ is a tree-decomposition of $G \otimes H$. Therefore :

Fact 3.1.

- $\text{twd}(G \otimes H) \leq \min \{ (\text{twd}(G) + 1) \cdot |V_H|, (\text{twd}(H) + 1) \cdot |V_G| \} - 1$.
- $\text{pwd}(G \otimes H) \leq \min \{ (\text{pwd}(G) + 1) \cdot |V_H|, (\text{pwd}(H) + 1) \cdot |V_G| \} - 1$.

In the following we prove a better upper bound for pathwidth, and under certain conditions, a better upper bound for treewidth. First, we give an example that will prove that the trivial bound from Fact 3.1i. cannot be improved for arbitrarily given graphs.

Proposition 3.2. *There exists an infinite set of pairs of graphs $\{(G_{k,p}, H_p), k \geq 2, p \geq 2\}$ such that, $|V_{H_p}| = p$, $\text{twd}(G_{k,p}) = k$, and $\text{twd}(G_{k,p} \otimes H_p) = (k + 1) \cdot p - 1$.*

Proof. The graph H_p is any connected graph on $p \geq 2$ vertices. The graph $G_{k,p}$ is obtained by considering a $(k + 1)$ -clique, K_{k+1} , with vertices denoted by $1, 2, \dots, k + 1$. For each i , $1 \leq i \leq k + 1$, we add $p(p - 1)/2$ new vertices denoted by $v_i(1), v_i(2), \dots, v_i(p(p - 1)/2)$ (Fig. 1), and we connect each of them to vertex i of K_{k+1} . For each ordered pair i, j , $1 \leq i, j \leq k + 1, i \neq j$, we add $p(p - 1)/2$ new vertices denoted by $e_{ij}(1), e_{ij}(2), \dots, e_{ij}(p(p - 1)/2)$; each of them is connected to both vertex i and vertex j . In the graph $G_{k,p} \otimes H_p$, we denote by $H^i(\alpha)$, $1 \leq i \leq k + 1, 1 \leq \alpha \leq p(p - 1)/2$, the copy of H_p with vertex set $\{(v_i(\alpha), 1), (v_i(\alpha), 2), \dots, (v_i(\alpha), p)\}$. In the graph $G_{k,p} \otimes H_p$, we also denote by $H^{ij}(\alpha)$, $1 \leq i, j \leq k + 1, i \neq j, 1 \leq \alpha \leq p(p - 1)/2$, the copy of H_p with vertex set $\{(e_{ij}(\alpha), 1), (e_{ij}(\alpha), 2), \dots, (e_{ij}(\alpha), p)\}$. The set $\{H^i(\alpha), H^{ij}(\alpha), 1 \leq i, j \leq k + 1, i \neq j, 1 \leq \alpha \leq p(p - 1)/2\}$, is a set of pairwise vertex-disjoint graphs. The graph $G_{k,p}$ is a partial k -tree and it has a clique on $k + 1$ vertices. So, its treewidth is k . In the following, we prove that the graph $G_{k,p} \otimes H_p$ has a subgraph homeomorphic to $K_{(k+1) \cdot p}$. More precisely, we prove that $G_{k,p} \otimes H_p$ contains a subdivision of $K_{(k+1) \cdot p}$, where the pairs of vertices connected by the paths of the subdivision are all the pairs of the form $(i, r), (j, s)$ where i (resp. j) is the vertex i (resp. j) in the copy of K_{k+1} contained in the r th (resp. s th) copy of $G_{k,p}$. First, consider all the $(p(p - 1)/2) \cdot (k + 1)$ pairs of vertices of the form $(i, r), (i, s)$ with $r < s$, and $i, 1 \leq i \leq k + 1$. Each of these pairs is connected by a path, say $P^i(\alpha)$, $1 \leq \alpha \leq p(p - 1)/2, 1 \leq i \leq k + 1$, with all of internal vertices contained in the copy $H^i(\alpha)$ of H_p . The set S_v of all those paths is a set of pairwise disjoint paths because their internal vertices are contained in pairwise disjoint copies of H_p . Now, consider the set I of all the $(p(p - 1)/2) \cdot k(k + 1)$ pairs of vertices of the form $(i, r), (j, s)$, with $r < s, 1 \leq i, j \leq k + 1, i \neq j$. For each ordered pair (i, j) , we use a path, say $P^{ij}(\alpha)$, in the copy $H^{ij}(\alpha)$ of H_p , $1 \leq \alpha \leq p(p - 1)/2$, to connect a different pair from I . Let S_e be the set of all those $p(p - 1) \cdot k(k + 1)/2$ paths. The set $S_e \cup S_v$ is a set of pairwise disjoint paths that constitute a subdivision of $K_{(k+1) \cdot p}$.

Since $G_k \otimes H_k$ has a subgraph homeomorphic to $K_{(k+1) \cdot p}$, it cannot have a treewidth less than $(k + 1) \cdot p - 1$. Hence the pair $(G_{k,p}, H_p)$ is as claimed. $\square$

Fact 3.3. *A tree on n vertices, $n \geq 2$, has a rooted fundamental tree-decomposition in which all bags are relax-introduce.*

Proof. We root the tree at a fixed leaf. In the notation $[u, v]$ used for the edges of the rooted tree, the extremity u is the parent of v . At each edge $[u, v]$ of the tree, we associate a bag $\{u, v\}$. We make each bag $\{u, v\}$ the parent of each bag $\{v, w\}$, where the associated edges are of the form $[u, v], [v, w]$ respectively. We obtain a tree-decomposition satisfying all the conditions in the statement. $\square$

Theorem 3.4. i. If G has a rooted quasi-fundamental tree-decomposition with no blocking bag then, for any graph H , $\text{twd}(G \otimes H) \leq \text{twd}(G) \cdot |V_H| + \text{twd}(H)$.
 ii. If G has a rooted quasi-fundamental tree-decomposition whose bags of size $\text{twd}(G) + 1$ are all relax-introduce or single-branch-leaving then, for any graph H , $\text{twd}(G \otimes H) \leq \text{twd}(G) \cdot |V_H|$.

Proof. We propose two procedures that, given a rooted irredundant tree-decomposition $(\mathcal{B}, (Y_r))$ of G with no blocking bag, and a rooted tree-decomposition $(\mathcal{A}, (X_s))$ of H , produce a tree-decomposition $(\mathcal{T}, (Z_t))$ of $G \otimes H$. Let P be a rooted path of $q = |V_H|$ nodes. The first procedure that we called **Construct**, processes the tree-decomposition of G in a prefix-order and decides, at each node, which type of compound by P or by $\mathcal{A}$ has to be applied according to the type of the corresponding bag.

At the end of this first procedure, one obtains a tree $\mathcal{T}$ which is the base of a tree-decomposition for the cartesian product. A second procedure, **Fill-bags**, processes this tree and decides, again according to the bag types of the given tree-decomposition of G , what is the content of the current bag.

We denote by $0, 1, \dots, q-1$, the nodes of P from the root down to the leaf. We denote the nodes of $\mathcal{A}$ by s_0, s_1 , etc., where s_0 is the root. We denote by $w(G)$ and $w(H)$ the widths of the given tree-decompositions of G and H respectively.

The structure T , initially set to $T := \mathcal{B}$, is updated by the procedure **Construct** in a prefix-order by examining the type of each bag in $(\mathcal{B}, (Y_r))$. The first call of the following procedure is performed with $T := \mathcal{B}$, and $r := r_0$, where r_0 is the root of $\mathcal{B}$.

```
Construct(( $\mathcal{B}, (Y_r)$ ),  $T, r, q, \mathcal{A}$ ) {
  begin
    If  $r$  is such that  $|Y_r| \leq w(G)$  then {no compound occurs at  $r$ }
      begin Rename  $r$  to  $(r, 0)$ ; end
    elseif  $Y_r$  is relax-introduce then
      begin  $T := T[r \downarrow P]_{(q-1)}$ ;
        {this renames the nodes of  $P$  to  $(r, 0), \dots, (r, (q-1))$ }.
      end
    elseif  $Y_r$  is  $r_\beta$ -single-branch-leaving then
      begin  $T := T[r \downarrow_{r_\beta} P]$ ;
        {this renames the nodes of  $P$  to  $(r, 0), \dots, (r, (q-1))$ }.
      end;
    else {this is furtive-introduce}
      begin  $T := T[r \downarrow \mathcal{A}]$  ;
        {this renames each node  $s_i$  of  $\mathcal{A}$  to  $(r, i)$ }.
      end;
    If  $r$  is not a leaf
      begin
        For each child  $r_\gamma$  of  $r$ 
          begin Construct(( $\mathcal{B}, (Y_r)$ ),  $T, r_\gamma, q, \mathcal{A}$ ) end;
        end;
      end;
  end; }
```

Let $\mathcal{T}$ be the tree T obtained at the end of the previous procedure. Every node t of $\mathcal{T}$ has a name of the form (r, i) . We say that t has prefix r . The root has the name $t_0 = (r_0, 0)$, where r_0 is the root of $\mathcal{B}$.

Notation. For every $r \in V_{\mathcal{B}}$, we denote by $\mathcal{T}_r$, the subtree of $\mathcal{T}$ induced by all the nodes t with prefix r .

The next procedure, given a tree $\mathcal{T}$ produced by application of the previous procedure, tells what is the content of each bag $Z_t, t \in V_{\mathcal{T}}$ according to the type of each bag of $(\mathcal{B}, (Y_r))$. The vertices of H are denoted by $1, \dots, q$. The first call is performed with $r := r_0$.

```
Fill-bags(( $\mathcal{B}, (Y_r)$ ),  $\mathcal{T}, r, q, (\mathcal{A}, (X_s))$ ) {
  begin
    If  $r$  is such that  $|Y_r| \leq w(G)$  then
      begin
         $Z_{(r,0)} = Y_r \times V_H$ ;
      end
    elseif  $Y_r$  is relax-introduce then
```

```

begin
  pick a synchronized couple (y, z);
  For i := 1 to q do
    begin
       $Z_{(r,q-i)} = (Y_r \setminus \{y, z\}) \times V_H \cup \{y\} \times \{i, \dots, q\} \cup \{z\} \times \{1, \dots, i\}$ ;
    end;
  end
  elseif  $Y_r$  is  $r_\beta$ -single-branch-leaving then
  begin
    pick a synchronized couple (y, x)
    For i := 1 to q do
      begin
         $Z_{(r,q-i)} = (Y_r \setminus \{x, y\}) \times V_H \cup \{y\} \times \{i, \dots, q\} \cup \{x\} \times \{1, \dots, i\}$ ;
      end;
    end
  else { $Y_r$  is furtive-introduce}
  begin
    pick a vertex y that is furtive;
    For i := 0 to  $|\mathcal{A}| - 1$  do
      begin
         $Z_{(r,i)} = (Y_r \setminus \{y\}) \times V_H \cup \{y\} \times X_{s_i}$ ;
      end;
    end;
  end;
  If r is not a leaf
  begin
    For each child  $r_\gamma$  of r
      begin Fill-bags( $(\mathcal{B}, (Y_r)), \mathcal{T}, r_\gamma, q, (\mathcal{A}, (X_s))$ ) end;
    end;
  end; }
end; }

```

Our construction has the following properties:

Property 3.5. For every r , every vertex y that is not selected at r as a furtive vertex satisfies:

- i. If y was selected at r as a leaving vertex then: $\forall t, t \in V_{\mathcal{T}_r}, \forall j, 1 \leq j \leq q$, if $(y, j) \in Z_t$ then $(y, j) \in Z_{t'}$, for all nodes t' on the path between t and the leaf $(r, q - 1)$ of the path $\mathcal{T}_r$. Moreover, $\{y\} \times V_H \subset Z_{(r,q-1)}$.
- ii. If y was selected at r as the partner of a leaving vertex then: $\forall t, t \in V_{\mathcal{T}_r}, \forall j, 1 \leq j \leq q$, if $(y, j) \in Z_t$ then $(y, j) \in Z_{t'}$, for all nodes t' on the path between the root $(r, 0)$ of the path $\mathcal{T}_r$ and t . Moreover, $\{y\} \times V_H \subset Z_{(r,0)}$.
- iii. If y was not selected at r then: $\forall t, t \in V_{\mathcal{T}_r}, \{y\} \times V_H \subset Z_t$.

Property 3.6. For every r , let G_r be the subgraph of G induced by all the vertices in Y_r . The subtree $\mathcal{T}_r$ of $\mathcal{T}$ is a tree-decomposition of the graph $G_r \otimes H$.

This decomposition is of width at most $w(G) \cdot |V_H| + w(H)$. It is of size at most $w(G) \cdot |V_H|$ if, in addition, Y_r is not furtive-introduce.

Proof (Checking Condition (P1) of the Definition of a Tree-decomposition). We want to prove that the union of all the Z_t 's where t has prefix r is $Y_r \times V_H$. By definition, $Z_t \subseteq Y_r \times V_H$, for all t with prefix r . For the other inclusion we note the following. If y was not selected at r , then by the last observation in [Property 3.5](#), $\{y\} \times V_H \subset Z_t$, for all t with prefix r . If Y_r is furtive-introduce, and y is the furtive vertex picked by the procedure, we have $(Y_r \setminus \{y\}) \times V_H \subset Z_t$, for all t with prefix r . Since, $(\mathcal{A}, (X_s))$ is a tree-decomposition of H , we have $\{y\} \times V_H = \bigcup_s \{y\} \times X_s$. Therefore, $\{y\} \times V_H \subset \bigcup_s Z_{(r,s)}$. If Y_r is not furtive-introduce, we use the observations in [Property 3.5](#) to deduce that, if a couple (y, z) was selected at r then $\{u\} \times V_H \subset Z_t$, for every $u \notin \{y, z\}$, and every t with prefix r , and that $\{y\} \times V_H \subset Z_{(r,q-1)}$, and $\{z\} \times V_H \subset Z_{(r,0)}$.

Checking Condition (P2) of the Definition of a Tree-decomposition.

Let e be an edge of $G_r \otimes H$. We want to check that it has both ends in some Z_t , where t has prefix r . Let $e = [(u, i), (u', i')]$ with $[i, i'] \in E_H$. By [Property 3.5](#), the only case that needs a check is that of u matching the furtive vertex picked by the procedure, in case Y_r is furtive-introduce. Since $(\mathcal{A}, (X_s))$ is a tree-decomposition for H , there exists s such that $[i, i'] \in X_s$. Therefore, $e \in Z_{(r,s)}$. Now, assume that e is of the form $[(u, i), (u', i)]$ with $[u, u'] \in E_{G_r}$. If Y_r is furtive-introduce, (u, i) and (u', i) are together in $Z_{(r,s)}$ where s is such that $i \in X_s$. Using the observations in [Property 3.5](#), the only remaining case that needs a check is that of u, u' involved in the synchronized couple in case a compound by the path P occurs at r . In this case, (u, i) and (u', i) are together in $Z_{(r,q-i)}$.

Checking Condition (P3) of the Definition of a Tree-decomposition.

If no compound occurs at r then $\mathcal{T}_r$ is reduced to a single node. Assume that a compound by the path P occurred at r , and let i, i', i'' such that $1 \leq i'' < i' < i \leq q$. We want to prove that $Z_{(r, q-i)} \cap Z_{(r, q-i'')} \subseteq Z_{(r, q-i')}$. Let (y, z) be the synchronized couple picked by the procedure. We have $Z_{(r, q-i)} \cap Z_{(r, q-i'')} = (Y_r \setminus \{y, z\}) \times V_H \cup \{y\} \times \{i, \dots, q\} \cup \{z\} \times \{1, \dots, i''\} \subseteq (Y_r \setminus \{y, z\}) \times V_H \cup \{y\} \times \{i', \dots, q\} \cup \{z\} \times \{1, \dots, i'\} = Z_{(r, q-i')}$. Finally, assume that a compound by the tree $\mathcal{A}$ occurred at r . Let y be the furtive vertex picked by the procedure, and let $(r, s), (r, s'), (r, s'')$ be nodes of $\mathcal{T}_r$ such that (r, s') is on the path between (r, s) and (r, s'') . By definition of the compound, $\mathcal{T}_r$ is isomorphic to $\mathcal{A}$, and s, s', s'' are nodes of $\mathcal{A}$, and s' is on the path between s, s'' . Since $(\mathcal{A}, (X_s))$ is a path-decomposition of H , we have $X_s \cap X_{s''} \subseteq X_{s'}$. Therefore, $Z_{(r, s)} \cap Z_{(r, s'')} = (Y_r \setminus \{y\}) \times V_H \cup \{y\} \times (X_s \cap X_{s''}) \subseteq (Y_r \setminus \{y\}) \times V_H \cup \{y\} \times X_{s'} = Z_{(r, s')}$.

If a compound by the path P occurred at r then for all t with prefix r , Z_t is of size exactly $w(G) \cdot |V_H| + 1$. If a compound by the tree $\mathcal{A}$ occurred at r then for all t with prefix r , Z_t is of size at most $w(G) \cdot |V_H| + w(H) + 1$. This ends the proof of [Property 3.6](#).

Claim. The decomposition $(\mathcal{T}, (Z_t))$ obtained at the end of the execution performed by calling `Fill-bags` $((\mathcal{B}, (Y_r)), \mathcal{T}, r_0, q, (\mathcal{A}, (X_s)))$ is a tree-decomposition of $G \otimes H$.

By construction of $(\mathcal{T}, (Z_t))$, $V_{\mathcal{T}} = \bigcup_{r \in V_{\mathcal{B}}} V_{\mathcal{T}_r}$, and every node t of $\mathcal{T}$ is of the form (r, i) . By [Property 3.6](#), $\bigcup_{(r, i) \in V_{\mathcal{T}}} Z_{(r, i)} = \bigcup_r Y_r \times V_H = (\bigcup_r Y_r) \times V_H = V_G \times V_H$. This proves that condition (P1) is satisfied. Since $(\mathcal{B}, (Y_r))$ is a tree-decomposition of G , every edge of $G \otimes H$ is an edge of $G_r \otimes H$ for some $r \in V_{\mathcal{B}}$. Then we use [Property 3.6](#), to prove that condition (P2) is also satisfied.

By construction of $\mathcal{T}$, if $Q[t_1, t_2]$ is the path of $\mathcal{T}$ between the nodes t_1 and t_2 then the subgraph of $\mathcal{B}$ induced by the set $\{r/t \text{ has prefix } r \text{ and } t \text{ is a node of } Q[t_1, t_2]\}$ is the path of $\mathcal{B}$ between the nodes r_1 and r_2 , where r_1, r_2 are the prefixes of t_1 and t_2 . We denote by *map* the corresponding mapping from the set of the paths of $\mathcal{T}$ to the set of the paths of $\mathcal{B}$. Also by construction, for every $r \in V_{\mathcal{B}}$, *map* $(Q[(r, i_1), (r, i_2)])$ is the path reduced to the single node r .

Let $t = (r, i), t' = (r', i'), t'' = (r'', i'')$ be three distinct nodes of $\mathcal{T}$ such that t' is on the path $Q[t'', t]$. We want to prove that $Z_t \cap Z_{t''} \subseteq Z_{t'}$. If t, t', t'' have the same prefix then we are finished ([Property 3.6](#)). So, we assume $r \neq r''$. Let a be the lowest common ancestor of r and r'' . We can assume, w.l.o.g., that $a \neq r$. Let $j, 1 \leq j \leq q$, and let $(y, j) \in Z_t \cap Z_{t''}$. Since, $(\mathcal{B}, (Y_r))$ is a tree-decomposition,

$$y \in Y_s, \text{ for all } s, s \in \text{map}(Q[t'', t']). \quad (1)$$

We define the following subsets of the set of nodes of *map* $(Q[t'', t'])$:

Notation.

$S(y) = \{s \text{ node of } \text{map}(Q[t'', t'])/y \text{ was selected at } s\};$

$S_f(y) = \{s \in S(y)/y \text{ is furtive at } s\};$

$S_l(y) = \{s \in S(y)/y \text{ is leaving at } s\};$

$S_{pc}(y) = \{s \in S(y)/y \text{ is partner for the leaving vertex in a compatible compound at } s\};$

$S_{ps}(y) = \{s \in S(y)/y \text{ is partner for the leaving vertex in an } s_{\beta}\text{-single-branch compound at } s\}.$

These are all the manners, a selection of y can be made at s . The [Definition 2.3](#), together with (1) imply that $S_f(y) = \emptyset$. Hence,

$$S(y) = S_l(y) \cup S_{pc}(y) \cup S_{ps}(y). \quad (2)$$

The [Definition 2.3](#), together with (1) also imply all the following: $S_{pc}(y) \cap S_{ps}(y) = \emptyset$ (otherwise, we would have the contradiction that for some $s, y \in Y_r \setminus d(s)$, and $y \in d(s)$), $S_l(y) \cap S_{pc}(y) = \emptyset$, $S_l(y) \cap S_{ps}(y) = \emptyset$, and, moreover:

$$S_l(y) \subseteq \{a\} \quad (3)$$

and,

$$S_{pc}(y) \subseteq \{r, r''\}, \text{ and } a \notin S_{pc}(y) \quad (4)$$

and,

$$s \in S_{ps}(y) \Rightarrow s_{\beta} \notin \text{map}(Q[t'', t]) \text{ and the root } (s, 0) \text{ of } \mathcal{T}_s \text{ is the unique node of } \mathcal{T}_s \text{ used by } Q[t'', t]. \quad (5)$$

Hence, using the second observation in [Property 3.5](#), we have:

$$\text{For every } s \in S_{ps}(y), \text{ if } t' \in Q[t'', t] \text{ has prefix } s \text{ then } \{y\} \times V_H \subset Z_{t'}. \quad (6)$$

Combining (3), (4), (6) and the two last observations in [Property 3.5](#), we obtain:

$$\forall s, s \neq a, s \in \text{map}(Q[t'', t]) - \{r, r''\} \Rightarrow \{y\} \times V_H \subset Z_{t'}, \text{ for all } t' \in Q[t'', t] \text{ with prefix } s; \quad (7)$$

and,

$$S_l(y) = \emptyset \Rightarrow \text{if } t' \in Q[t'', t] \text{ has prefix } a \text{ then } \{y\} \times V_H \subset Z_{t'}. \quad (8)$$

Combining (3), (4), (6) and all the observations in [Property 3.5](#), we obtain:

$$(y, j) \in Z_{t'}, \text{ for all } t' \in Q[t'', t] \text{ with prefix in } \{r, r''\} - \{a\}. \quad (9)$$

We discuss two cases according to $S_l(y)$.

- $S_l(y) = \emptyset$.

Using (7) and (8), we have: $\{y\} \times V_H \subset Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix not in $\{r, r''\}$ or with prefix a ; and using (9), we obtain that $(y, j) \in Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix in $\{r, r''\}$.

- $S_l(y) = \{a\}$.

If the compound at a is a_β -single-branch then $a = r''$, and the a_β -child is on the path $Q[t'', t]$ (possibly $r = a_\beta$). In this case, $(a_\beta, 0)$ and its parent $(a, q - 1)$ are nodes of $Q[t'', t]$. Moreover, the subpath of the path $\mathcal{T}_a$ between t'' and the leaf $(a, q - 1)$ of $\mathcal{T}_a$ is a subpath of $Q[t'', t]$. By the first observation in [Property 3.5](#), we have $(y, j) \in Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix a . Using (7), we obtain that $\{y\} \times V_H \subset Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix not in $\{r, a\}$, and using (9) we obtain that $(y, j) \in Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix r .

Suppose that the compound at a is compatible. If $a = r''$, the subpath of the path $\mathcal{T}_a$ between t'' and the leaf $(a, q - 1)$ of $\mathcal{T}_a$ is a subpath of $Q[t'', t]$. If $a \neq r''$, the leaf $(a, q - 1)$ of $\mathcal{T}_a$ is the unique node of $\mathcal{T}_a$ used by the path $Q[t'', t]$. By the first observation in [Property 3.5](#), we obtain that $(y, j) \in Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix a . By (7), we obtain that $\{y\} \times V_H \subset Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix not in $\{r, r''\}$ and not a , and by (9), we obtain that $(y, j) \in Z_{t'}$, for all t' on the path $Q[t'', t]$ with prefix in $\{r, r''\}$.

This ends the proof of the claim and the proof of [Theorem 3.4](#). $\square$

The following corollary improves the bound in [Fact 3.1ii](#).

Corollary 3.7. *For any graphs G and H ,*

$$\text{pwd}(G \otimes H) \leq \min \{ \text{pwd}(G) \cdot |V_H| + \text{pwd}(H), \text{pwd}(H) \cdot |V_G| + \text{pwd}(G) \}.$$

Proof. From the first part of vii. of [Remark 2.4](#), our construction works without any condition if we consider an irredundant path-decomposition $(\mathcal{B}, (Y_r))$ of G and a path-decomposition $(\mathcal{A}, (X_s))$ of H and we apply at each node r such that Y_r has the maximum bag size, either a compound by the path P or by the path $\mathcal{A}$, according to whether we are at a relax-introduce bag or not. $\square$

The following corollary gives infinite classes of graphs for which the bound in [Fact 3.1i](#) is improved.

Corollary 3.8.

- If G is a tree (resp. a path) then, for any graph H ,

$$\text{twd}(G \otimes H) \leq |V_H| \quad (\text{resp. } \text{pwd}(G \otimes H) \leq |V_H|).$$

- Let H be a fixed graph that has a hamiltonian path. If $\mathcal{K}$ is the class of all trees having a path of length at least $|V_H| - 1$ then,

$$\{G \otimes H, G \in \mathcal{K}\} \subset \text{TWD} (= |V_H|).$$

- For any graphs G and H such that $\text{pwd}(G) = \text{twd}(G)$,

$$\text{twd}(G \otimes H) \leq \text{twd}(G) \cdot |V_H| + \text{twd}(H).$$

Proof. The first assertion is immediate from [Fact 3.3](#) and [Theorem 3.4ii](#). The second one is due to the fact that, $G \otimes H$ has a subgraph isomorphic to the square grid of dimension $|V_H|$. The last assertion is immediate from the second part of vii of [Remark 2.4](#). $\square$

Question: Can one obtain a characterization for the graphs having a quasi-fundamental tree-decomposition with no blocking bag?

4. Equational and definable sets of graph products

4.1. HR-sets of graph products

If $\mathcal{G}$ and $\mathcal{H}$ are graph classes, we denote by $\otimes_{\mathcal{G}}(\mathcal{H})$ the set of graphs $\{G \otimes H, G \in \mathcal{G}, H \in \mathcal{H}\}$.

Theorem 4.1. Let $\mathcal{G}$ be a finite set of graphs and $\mathcal{I}$ be a class of graphs. If $\mathcal{I}$ is HR-equational then so is $\otimes_{\mathcal{G}}(\mathcal{I})$.

Proof. For each graph H in $\mathcal{G}$ we consider a set $\mathcal{K}_H = \{1_H, 2_H, \dots, q_H\}$ of $|V_H|$ source labels.

Let M (resp., $\otimes_{\mathcal{G}}M$) be the algebra of all graphs with source labels in a countable set $\mathcal{C}$ (resp., $\mathcal{C} \times \cup_{H \in \mathcal{G}} \mathcal{K}_H$). For each $H \in \mathcal{G}$, we enlarge the set of constant symbols of $\otimes_{\mathcal{G}}M$ by all the graphs $\otimes_H c$, $c \in \mathcal{C}$, $H \in \mathcal{G}$, denoting the graph H where every vertex i_H is the (c, i_H) -source, and all the graphs $\otimes_H cc'$, $c, c' \in \mathcal{C}$, $H \in \mathcal{G}$, obtained from the cartesian product of H by an edge and labeling in a copy of H , each vertex i_H by (c, i_H) and in the other copy each vertex i_H by (c', i_H) .

Let us define, for each $H \in \mathcal{G}$, a mapping $f_H : M \rightarrow \otimes_{\mathcal{G}}M$, inductively on the form of the terms of M . This mapping is used to transform the polynomial system from which $\mathcal{I}$ is obtained to a polynomial system S_H for which a component has $\otimes_{\{H\}}(\mathcal{I})$ as the least solution. If t is a constant symbol then, we let $f_H(t) = \otimes_H t$. If $t = r //_{C, C'} s$ then, $f_H(t) = f_H(r) //_{C \times \mathcal{K}_H, C' \times \mathcal{K}_H} f_H(s)$.

Finally, if $t = fg_c(r)$ then, $f_H(t) = fg_{(c, 1_H)}(fg_{(c, 2_H)}(\dots(fg_{(c, q_H)}(f_H(r))) \dots))$.

There exists a finite subset $\mathcal{K}$ of $\mathcal{C}$ such that $\mathcal{I}$ is obtained as a component of the least solution of a polynomial system S over $F_{HR}(\mathcal{K})$. Let $S = \langle u_1 = p_1, \dots, u_n = p_n \rangle$, and assume w.l.o.g., that $\mathcal{I} = L(S, M, u_1)$.

Let $S_H = \langle u_1 = \otimes_H p_1, \dots, u_n = \otimes_H p_n \rangle$ be the polynomial system over $F_{HR}(\mathcal{K} \times \mathcal{K}_H)$ where each $\otimes_H p_j$ is obtained from p_j by replacing each monomial t_α of p_j by $f_H(t_\alpha)$.

Note that, in each S_H the sort of u_1 (in the algebra $\otimes_{\mathcal{G}}M$) is $C \times \mathcal{K}_H$ where C is the sort (in the algebra M) of the unknown u_1 of S .

Let us denote by $fg_{C \otimes \mathcal{K}_H}$ the composition of mappings $\circ_{c \in C} \{fg_{(c, 1_H)} \circ fg_{(c, 2_H)} \circ \dots \circ fg_{(c, q_H)}\}$, and consider the monomial: $p_0^H = fg_{C \otimes \mathcal{K}_H}(u_1)$. It is of the sort: $\emptyset$.

Consider any sequence of the systems S_H and rename the unknowns of the second system in the sequence to $u_{n+1}, \dots, u_{2n}$ respectively, and the unknowns of the third system to $u_{2n+1}, \dots, u_{3n}$ respectively, etc.

Now, let S' be the polynomial system over $F_{HR}(\mathcal{K} \times \cup_{H \in \mathcal{G}} \mathcal{K}_H)$ obtained as follows. First, merge all the equations of all the systems S_H . Then, add the equation $u_0 = \bigoplus_{H \in \mathcal{G}} p_0^H$.

We have, $\otimes_{\mathcal{G}}(\mathcal{I}) = L((S', \otimes_{\mathcal{G}}M), u_0)$. $\square$

4.2. MS-definability of graph product sets

4.2.1. Graphs as relational structures

As in [4], if G is a graph, we denote by $[G]$ the relational structure associated with G with the set of vertices as the domain of the structure, and the binary relation adj_G of adjacency. We also denote by $[G]$ the relational structure associated to G with the set of vertices and edges as the domain of the structure, and the binary relation inc_G defined by $\text{inc}_G(x, y)$ iff x is a vertex of G , y is an edge of G and x is incident with y . In the sequel, we generally omit the subscript in the notation of the adjacency and incidence relations. We denote by MS_1 and MS_2 , the monadic second order logic for $[G]$ structures and $[G]$ structures respectively.

4.2.2. Formulas for substructures

Let $\mathcal{R}$ be a relational signature and S be an $\mathcal{R}$ -structure. If C is a subset of the domain of S , the *substructure* $S[C]$ of S is the structure with domain C and with relations $R_{S[C]} = R_S \cap C^{\rho(R)}$, for all $R \in \mathcal{R}$ and $\rho(R)$ denoting the arity of R .

Let G be a graph and let X be a nonempty subset of $V_G \cup E_G$.

We let $X_v = \{v \in V_G / v \text{ is incident with an edge from } X\}$, $X_e = \{e \in E_G / e \text{ has both ends in } X\}$, $\overline{X}_v = (X \cap V_G) \cup X_v$, $\overline{X}_e = (X \cap E_G) \cup X_e$ if X is disjoint from V_G or from E_G , and $\overline{X}_e = (X \cap E_G) \cap X_e$ otherwise. We let $\overline{X} = \overline{X}_v \cup \overline{X}_e$. We denote by G_X , the graph with vertex set $X \cap V_G$ and with edge set $\overline{X} \cap E_G$. We say that G_X is the subgraph of G induced by X .

The substructure of the structure $[G]$ representing G_X , where $X \subseteq V_G$, is thus $[G_X] = [G][X]$, and the substructure of the structure $[G]$ representing G_X , where $X \subseteq V_G \cup E_G$, is $[G_X] = [G][\overline{X}]$.

An *MS-formula* $\varphi(x_1, \dots, x_n, X_1, \dots, X_m)$ is a monadic second order formula possibly with free variables. If φ has first order variables then they are in $\{x_1, \dots, x_n\}$, and if φ has second order variables then they are in $\{X_1, \dots, X_m\}$. The following lemma holds for second order formulas (see, [4]). We give a statement of the lemma for *MS-formulas* only.

Lemma 4.2. Let $\varphi(x_1, \dots, x_n, X_1, \dots, X_m)$ be an *MS-formula* for $\mathcal{R}$ -structures; one can construct an *MS-formula* $\varphi'(x_1, \dots, x_n, X_1, \dots, X_m, X)$ such that the following holds for every $\mathcal{R}$ -structure S . For every subset D of the domain of S , for every $\bar{a} = (a_1, \dots, a_n) \in D^n$, for every $\bar{A} = (A_1, \dots, A_m) \in \mathcal{P}(D)^m$:

$$S \models \varphi'(\bar{a}, \bar{A}, D) \text{ iff } S[D] \models \varphi(\bar{a}, \bar{A}).$$

Sketch of the proof: We can assume that $X \notin \{X_1, \dots, X_m\}$, and if X is bound in φ we rename its bound occurrences. Let φ' be associated to φ by the following inductive definition:

$$\begin{aligned} (\forall Y. \psi)' &= \forall Y. \psi', (\exists Y. \psi)' = \exists Y. \psi', \\ (\forall x \psi)' &= \forall x[x \in X \Rightarrow \psi'], (\exists x \psi)' = \exists x[x \in X \wedge \psi'], \\ (\psi_1 \text{ op } \psi_2)' &= \psi_1' \text{ op } \psi_2', \text{ where } \text{op} \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}, (\neg \psi)' = \neg \psi', \\ (\psi)' &= \psi \text{ for every atomic formula } \psi. \end{aligned}$$

The formula φ' is called the relativization of φ to X and will be denoted by $\varphi_{[X]}$. Its quantifier height is the same as that of φ .

4.2.3. Notation for some basic formulas

Variables in uppercase letters denote set variables.

We write “ $x \neq x'$ ” instead of “ $\neg x = x'$ ”, “ $X(x)$ ” instead of “ $x \in X$ ”, and “ $X \subseteq X'$ ” instead of “ $\forall x [x \in X \Rightarrow x \in X']$ ”.

The notation x, x' is used to denote vertices and the notation X, X' is used to denote sets of vertices. In the same way, y, y' denote edges and Y, Y' denote sets of edges.

The following formulas are all FO in $[G]$ (set variables can occur only in atomic formulas of the form $X(x)$).

- $\text{Adj}(X, X')$ holds iff an adjacency exists between a vertex from X and a vertex from X' .
- $\text{Isom}_1(X, X')$ holds iff every vertex from X is adjacent to a vertex from X' , and every vertex from X' is adjacent to a vertex from X , and two distinct vertices x_1, x_2 from X are adjacent to two distinct vertices x'_1, x'_2 from X' such that $\text{adj}(x_1, x_2)$ iff $\text{adj}(x'_1, x'_2)$.

The following formulas are all FO in $[G]$:

- $\text{edge}(x, x')$ holds iff $x \neq x'$ and there exists an edge y such that $\text{inc}(x, y) \wedge \text{inc}(x', y)$.
- $\text{endsof}(x, x', y)$ holds iff the ends of the edge y are x, x' .
- $\text{EDG}(X, X')$ holds iff an edge exists between a vertex from X and a vertex from X' .
- $\text{Isom}_2(X, X')$ holds iff every vertex from X is connected by an edge to a vertex from X' , and every vertex from X' is connected by an edge to a vertex from X , and two distinct vertices x_1, x_2 from X are connected by edges to two distinct vertices x'_1, x'_2 from X' such that $\text{edge}(x_1, x_2)$ iff $\text{edge}(x'_1, x'_2)$.

4.2.4. Shortcuts for some useful formulas

We recall the following result which we shall use in the proof of our next theorem.

Lemma 4.3. • There exists an MS-formula expressing in $[G]$ the following property of vertices x, x' and a set X of vertices for an arbitrary graph G :

P1: $x \neq x'$ and x, x' are connected by a path with internal nodes all in X .

- There exists an MS-formula expressing in $[G]$ the following property of vertices x, x' and a set X (resp. Y) of vertices (resp. edges) for an arbitrary graph G :

P2: $x \neq x'$, and X (resp. Y) is the set of vertices (resp. edges) forming a path from x to x' .

- The following property of a nonempty subset X of vertices can be expressed in $[G]$ and in $[G]$ by an MS-formula:

P3: X is a connected component of G .

A proof can be found in [9,4], for example.

Notation. Let $\text{Reach}(x, x', X)$ and $\text{CC}(X)$ be formulas corresponding to the properties P1 and P3 respectively. Let $\text{Path}(x, x', X)$ (resp. $\text{Path}(x, x', Y)$) be a formula corresponding to the property P2.

Lemma 4.4. For a fixed integer q , the following property of vertices x and x' and nonempty pairwise disjoint sets $X_1, \dots, X_q$, of vertices can be expressed in $[G]$ and in $[G]$ by an MS-formula:

P_q : x and x' are connected by a path using only edges with ends in different X_i 's.

Proof. Such a formula can be:

$$\exists X \subseteq \bigcup_{i=1}^{i=q} X_i \wedge \text{Reach}(x, x', X) \wedge \forall x_1 \forall x_2 \left[X(x_1) \wedge X(x_2) \wedge \text{adj}(x_1, x_2) \Rightarrow \bigwedge_{i=1}^{i=q} \neg (X_i(x_1) \wedge X_i(x_2)) \right],$$

in $[G]$, and in $[G]$ it can be:

$$\exists X \subseteq \bigcup_{i=1}^{i=q} X_i \wedge \text{Path}(x, x', X) \wedge \forall x_1 \forall x_2 \left[X(x_1) \wedge X(x_2) \wedge \text{edge}(x_1, x_2) \Rightarrow \bigwedge_{i=1}^{i=q} \neg (X_i(x_1) \wedge X_i(x_2)) \right]. \quad \square$$

Notation. We denote by $\text{Alternateends}_q(x, x', X_1, \dots, X_q)$ a formula corresponding to the property P_q , above.

4.2.5. Preservation of MS-definability by the cartesian product

Theorem 4.5. Let $\mathcal{I}$ be a (not necessarily finite) class of simple graphs.

- i. If $\mathcal{G}$ is a finite set of simple connected graphs and $\mathcal{I}$ is MS_1 -definable then $\otimes_{\mathcal{G}}(\mathcal{I})$ is MS_1 -definable.
- ii. If $\mathcal{G}$ is a finite set of connected graphs (not necessarily simple) and $\mathcal{I}$ is MS_2 -definable then $\otimes_{\mathcal{G}}(\mathcal{I})$ is MS_2 -definable.
- iii. If $\mathcal{I}$ and $\mathcal{G}$ are MS_2 -definable classes of simple connected graphs then $\otimes_{\mathcal{G}}(\mathcal{I})$ is MS_2 -definable.

Proof: If X, X' are two disjoint subsets of V_G , we denote by $V(X, X')$ the set of adjacent pairs (z, z') of vertices with $z \in X$ and $z' \in X'$. We say that $V(X, X')$ defines an isomorphism from G_X onto $G_{X'}$ iff for every $z \in X$, there exists z' such that $(z, z') \in V(X, X')$, for every $z' \in X'$, there exists z such that $(z, z') \in V(X, X')$, and for every pair z_1, z_2 of distinct vertices from X , there exist distinct vertices z'_1, z'_2 from X' such that $(z_1, z'_1) \in V(X, X')$, $(z_2, z'_2) \in V(X, X')$, and z_1, z_2 are adjacent iff z'_1, z'_2 are adjacent.

- i. Let φ be an MS_1 -formula defining $\mathcal{I}$. We prove that, given a fixed simple connected graph H , with q vertices, the property:

P_H : “ G is a cartesian product of a graph from $\mathcal{I}$ by the graph H ”, is expressible in $[G]$ by an MS formula:

There exists a partition $X_1, X_2, \dots, X_q$, of the set of vertices of G such that all the following is true:

(a) $\bigwedge_{1 \leq i \leq q} [G] \models \varphi_{[X_i]}$.

(b) For all pairs i, j ,

An adjacency between a vertex from X_i and a vertex from X_j exists iff vertex i and vertex j are adjacent in H , and

if an adjacency exists between a vertex from X_i and a vertex from X_j then $V(X_i, X_j)$ defines an isomorphism from G_{X_i} onto G_{X_j} ,

(c) For every $i, 1 \leq i \leq q$, for every pair x, x' of vertices in X_i , if x and x' are connected by a path using only edges with ends in different X_j 's, then $x = x'$.

Let H be a fixed simple connected graph and let $q = |V_H|$.

First we prove that if G is of the form $G'' \otimes H$ for some $G'' \in \mathcal{I}$ then such a partition exists. For every $i \in V_H$, let X_i be the set of vertices of V_G of the form (u, i) , $u \in V_{G''}$. The set $\{X_1, \dots, X_q\}$ form a partition of V_G . By the definition of the cartesian product of graphs, each subgraph G_{X_i} is isomorphic to G'' . Hence, for every i , $[G] \models \varphi_{[X_i]}$. Also by the definition of the cartesian product of graphs, for every pair i, j of adjacent vertices of H , the set $\{(u, i), (u, j)\}, u \in V_{G''}\}$ is exactly the set $V(X_i, X_j)$, and we have, also by the definition of the cartesian product of graphs: for every u_1, u_2 from $V_{G''}$, $(u_1, i), (u_2, i)$ are adjacent iff $(u_1, j), (u_2, j)$ are adjacent. Hence, $V(X_i, X_j)$ defines an isomorphism. By the definition of the cartesian product of graphs, if a path in G is formed only by edges with ends in different X_{α} 's, then all the vertices of this path are of the form (u, k) for some fixed $u \in V_{G''}$.

Now, we prove that the existence of such a partition implies that G is of the form $G'' \otimes H$ for some $G'' \in \mathcal{I}$. Let G'' be (a copy of) the graph isomorphic to the subgraph of G induced by X_1 , and let $\sigma_1 : V_{G''} \rightarrow X_1$, be the corresponding isomorphism. We define the following subsets of vertices of H :

$S_1 = \{1\}$, and $S_{l+1} = \{i, i \in V_H - S_l, \text{ and } i \text{ is neighbor of some vertex in } S_l\}$. Let p be the greatest integer l such that $S_l \neq \emptyset$. Since H is connected, the union of all the S_l 's is V_H .

We can assume $p \geq 2$ (otherwise H is reduced to a single vertex). Using an induction on l , $1 \leq l \leq p-1$, we observe that condition (b) implies that all the X_j 's have the same cardinality $|V_{G''}|$. For every $i \in S_2$, we let $f_i : X_1 \rightarrow X_i$, be the isomorphism from the graph G_{X_1} onto the graph G_{X_i} defined by $V(X_1, X_i)$. Let $\sigma_i : V_{G''} \rightarrow X_i$, be the isomorphism from the graph G'' onto the graph G_{X_i} defined by $\sigma_i = f_i \circ \sigma_1$. Note that we have $V(X_1, X_i) = \{(\sigma_1(u), \sigma_i(u)), u \in V_{G''}\}$.

Fact 1. If there are adjacent vertices i, i' in S_2 , then the isomorphism defined by $V(X_i, X_{i'})$ is compatible with both σ_i and $\sigma_{i'}$; that is $V(X_i, X_{i'}) = \{(\sigma_i(w), \sigma_{i'}(w)), w \in V_{G''}\}$.

Proof of Fact 1:

Since $V(X_i, X_{i'})$ defines an isomorphism, we have $|V(X_i, X_{i'})| = |X_i| = |X_{i'}| = |V_{G''}|$. Hence, $V(X_i, X_{i'})$ and $\{(\sigma_i(w), \sigma_{i'}(w)), w \in V_{G''}\}$ have the same cardinality. We prove that $V(X_i, X_{i'}) \subseteq \{(\sigma_i(w), \sigma_{i'}(w)), w \in V_{G''}\}$. Let $(z, z') \in V(X_i, X_{i'})$ and let $u = \sigma_i^{-1}(z)$. If $z' \neq \sigma_{i'}(u)$ we would have a path: $\sigma_{i'}(u) - \sigma_1(u) - \sigma_i(u) = z - z'$ connecting two distinct vertices from $X_{i'}$ and this path has the property that each of its edges has ends from different X_{α} 's. This contradicts condition (c). This ends the proof of Fact 1.

For every l , $1 \leq l \leq p$, we denote by H_l the subgraph of H induced by $\bigcup_{m=1}^l S_m$, and we set $Z_l = \bigcup_{i \in V_{H_l}} X_i$. We denote by G_l the subgraph of G induced by Z_l . We observe that:

$$\forall l, 1 \leq l \leq p, |Z_l| = \sum_{i \in V_{H_l}} |X_i| = |V_{G''}| \cdot |V_{H_l}|. \quad (10)$$

Fact 2. For every l , $1 \leq l \leq p$, there exist graph isomorphisms, $\sigma_i : V_{G''} \rightarrow X_i$, $i \in V_{H_l}$ such that $g_l : V_{G'' \otimes H_l} \rightarrow V_{G_l}$ defined by $g_l((u, i)) = \sigma_i(u)$, for every $(u, i) \in V_{G'' \otimes H_l}$, is an isomorphism from the graph $G'' \otimes H_l$ onto the graph G_l .

Proof of Fact 2: By induction on l .

For every $i \in S_1 \cup S_2$, we consider the graph isomorphisms σ_i defined above. By the definition of σ_1 , the statement holds for $l = 1$. By the definition of the σ_i , $i \in V_{H_2}$, we have $Z_2 = \bigcup_{i \in V_{H_2}} \sigma_i(V_{G''})$. Hence g_2 is surjective. Recalling Eq. (10), we see that g_2 is a bijection. By condition (b) together with Fact 1., we have $\sigma_i(u)$ and $\sigma_{i'}(u')$ are adjacent in G_2 iff $(i = i'$ and u, u' are adjacent in G'') or $(i, i'$ are adjacent in H_2 and $u = u')$. This is equivalent to say that (u, i) and (u', i') are adjacent in $G'' \otimes H_2$. Therefore, the statement also holds for $l = 2$. Let $l, 3 \leq l \leq p$ be such that the statement holds for $l - 1$. Let $j \in S_l$ and let i be a neighbor of j in S_{l-1} . By condition (b), $V(X_i, X_j)$ defines an isomorphism. Let $h_{ij} : X_i \rightarrow X_j$, be the corresponding isomorphism from the graph G_{X_i} onto the graph G_{X_j} , and let $f_{ij} = h_{ij} \circ \sigma_i$. We want to prove that if i and i' are two distinct neighbors of j in S_{l-1} then $\forall u \in V_{G''}$, $f_{ij}(u) = f_{i'j}(u)$. Assume that there exists $u \in V_{G''}$ such that $f_{ij}(u) \neq f_{i'j}(u)$. Let $x = f_{ij}(u)$ and $x' = f_{i'j}(u)$. Since H_{l-1} is connected, we consider a path P , connecting i and i' in H_{l-1} . By the induction hypothesis, the graph isomorphisms $\sigma_i, i \in V_{H_{l-1}}$ are such that g_{l-1} is a graph isomorphism. Hence, the set $\{\sigma_i(u), i \in V_P\}$ is the set of vertices of a path of G_{l-1} connecting $\sigma_i(u)$ and $\sigma_{i'}(u)$, namely the path $P' = g_{l-1}(\{u\} \otimes P)$. The path P' extended by the edges $(\sigma_i(u), x)$ and $(\sigma_{i'}(u), x')$ is a path of G_l connecting two distinct vertices from X_j and formed only by edges with ends in different X_α 's. This contradicts condition (c). Hence, for every $j \in S_l$, for every $u \in V_{G''}$, $f_{ij}(u)$ is invariant when i ranges over the set of the neighbors of j in S_{l-1} . For every, $j \in S_l$, we let $\sigma_j = f_{ij}$, where i is any neighbor of j in S_{l-1} . We note that this defines $V(X_i, X_j)$ as exactly the set $\{(\sigma_i(u), \sigma_j(u)), u \in V_{G''}\}$. Let j, j' be two adjacent vertices in S_l . We want to prove that the set $V(X_j, X_{j'})$ defined by condition (b) is exactly the set $\{(\sigma_j(u), \sigma_{j'}(u)), u \in V_{G''}\}$. Since the two sets have the same cardinality, it is sufficient to prove that $V(X_j, X_{j'}) \subseteq \{(\sigma_j(u), \sigma_{j'}(u)), u \in V_{G''}\}$. Let $(z, z') \in V(X_j, X_{j'})$. We have $z = \sigma_j(u)$ for some $u \in V_{G''}$. Assume that $z' \neq \sigma_{j'}(u)$. Let i, i' in S_{l-1} be neighbors of respectively j and j' (possibly $i = i'$). Since H_{l-1} is connected, we consider a path P connecting i and i' in H_{l-1} . With the same arguments as above the path $P' = g_{l-1}(\{u\} \otimes P)$ is a path in G_{l-1} connecting $\sigma_i(u)$ and $\sigma_{i'}(u)$. Extending P' by $\sigma_i(u) - z - z'$ and $\sigma_{i'}(u) - \sigma_{j'}(u)$ gives a path connecting two distinct vertices from $X_{j'}$ and formed only by edges with ends in different X_α 's. This contradicts condition (c).

Let $g_l : V_{G'' \otimes H_l} \rightarrow V_{G_l}$, defined by $g_l(u, i) = \sigma_i(u)$, for every $(u, i) \in V_{G'' \otimes H_l}$. By definition of the σ_i , $i \in V_{H_l}$, we have $Z_l = \bigcup_{i \in V_{H_l}} \sigma_i(V_{G''})$. By the same arguments as used above for g_2 , we see that g_l is a bijection. Let $\sigma_i(u)$ and $\sigma_{i'}(u')$ in V_{G_l} . If i and i' are both in H_{l-1} , then observing that $g_l|_{V_{G'' \otimes H_{l-1}}} = g_{l-1}$, and using the induction hypothesis, we deduce that $\sigma_i(u)$ and $\sigma_{i'}(u')$ are adjacent in G_{l-1} (hence in G_l) iff $(u, i), (u', i')$ are adjacent in $V_{G'' \otimes H_{l-1}}$ (hence in $V_{G'' \otimes H_l}$). Let (z, z') be a pair of adjacent vertices of G_l with z, z' not both in G_{l-1} . Then, (z, z') is either of the form $(\sigma_j(u), \sigma_{j'}(u'))$ for some $j \in S_l$ and u, u' adjacent in G'' , or of the form $(\sigma_i(u), \sigma_j(u))$ for some $u \in V_{G''}$, and some $(i, j) \in (S_{l-1} \cup S_l) \times S_l$, with i, j adjacent in H_l . Both cases are equivalent to say that $g_l^{-1}(z), g_l^{-1}(z')$ are adjacent in $G'' \otimes H_l$. This ends the proof of Fact 2.

Therefore, G satisfies P_H iff a partition such described in the statement of the theorem exists.

The condition (b) can be expressed by the formula:

$$\bigwedge_{i=1}^q \bigwedge_{j=i+1}^q [\text{ADJ}(X_i, X_j) \Leftrightarrow \text{adj}_H(i, j)] \wedge (\text{ADJ}(X_i, X_j) \Rightarrow \text{Isom}_1(X_i, X_j));$$

and the condition (c) by the formula:

$$\bigwedge_{i=1}^q \forall x \forall x' [(X_i(x) \wedge X_i(x') \wedge \text{Alternateends}_q(x, x', X_1, \dots, X_q)) \Rightarrow x = x'].$$

Hence, there exists an MS_1 -formula, ϕ_H expressing the property P_H , and we have: $G \in \otimes_{\mathcal{J}}(\mathcal{I})$ iff $[G] \models \bigvee_{H \in \mathcal{J}} \phi_H$.

- ii. We assume that $\mathcal{I}$ is definable by means of the MS_2 -formula φ . The set $\mathcal{J}$ is given together with the constant $w_{\mathcal{J}}$, the maximum of the edge multiplicities of the graphs in $\mathcal{J}$. In the sequel, we omit the subscript in the latter notation.

Let $w \geq 2$. For every $r, 2 \leq r \leq w$, and every pair u, v of vertices, we consider the property: “there exist r pairwise distinct edges, each with ends u and v ”. This property is expressible, for example, by the formula: $\pi_r(u, v) = \exists y_1 \dots \exists y_r [\bigwedge_{i=1}^r \bigwedge_{j=i+1}^r y_i \neq y_j \wedge \text{endsof}(u, v, y_i)]$.

We can express the property of adjacent vertices u, v of an arbitrary graph from $\mathcal{J}$: “ u and v are connected by an edge with the multiplicity m ” by the formula $\varpi_m(u, v)$, with $\varpi_1(u, v) = \neg \pi_2(u, v)$, $\varpi_m(u, v) = \pi_m(u, v) \wedge \neg \pi_{m+1}(u, v)$, if $2 \leq m \leq w - 1$, and $\varpi_w(u, v) = \pi_w(u, v)$.

P_H : “ G is a cartesian product of a graph from $\mathcal{I}$ by the graph H ”, is expressible in $[G]$ by an MS -formula:

There exists a partition $X_1, X_2, \dots, X_q$, of the set of vertices of G such that all the following is true:

- $\bigwedge_{1 \leq i \leq q} [G] \models \varphi|_{X_i}$.
- For all pairs i, j ,
 - An edge between X_i and X_j exists iff there is an edge in H between i and j , and
 If there exists an edge in H between i and j then:
 - $V(X_i, X_j)$ defines an isomorphism, and
 - For every $u \in X_i$, for every $v \in X_j$, if there is an edge between u and v , then this edge has multiplicity m iff the edge between i and j has multiplicity m .
- For every $i, 1 \leq i \leq q$, for every pair x, x' of vertices from X_i , if x and x' are connected by a path using only edges with ends in different X_j 's, then $x = x'$.

The first sub-condition in condition (b) can be expressed by: $\text{EDG}(X_i, X_j) \Leftrightarrow \text{edge}_H(i, j)$. The condition (i) can be expressed by: $\text{Isom}_2(X_i, X_j)$. The condition (ii) can be expressed by:

$$\forall u \forall v [X_i(u) \wedge X_j(v) \wedge \text{edge}(u, v) \Rightarrow \bigvee_{m=1}^{m=w} (\varpi_m(u, v) \Leftrightarrow \varpi_m(i, j))].$$

Condition (c) can be expressed as done for the condition (c) in the first part of the proof by using for $\text{Alternateends}_q(x, x', X_1, \dots, X_q)$ the version in the proof of [Lemma 4.4](#) given for $\lceil G \rceil$ structures.

iii. Let φ and φ' be MS_2 -formulas defining $\mathcal{I}$ and $\mathcal{J}$ respectively. We consider the following property of a graph G :

G is connected,

and

There exists a partition of the edges of G into two sets Y, Y' , such that all the following is true:

- (a) For each connected component X of the subgraph induced by Y , the relativization of φ to $\bar{X}$ holds in G ,
- (b) For every pair X, X' of connected components of the subgraph induced by Y , either there is no edge between X and X' , or $V(X, X')$ defines an isomorphism,
- (c) For each connected component X of the subgraph induced by Y , for every pair x, x' of vertices from X , if x and x' are connected by a path using only edges with ends in different connected components of the subgraph induced by Y , then $x = x'$.
- (d) The analogues of (a), (b), (c) obtained by replacing Y by Y' , and φ by φ' .

Let us denote by $\text{Alternateends}(x, x', Y)$ a formula saying that “ x and x' are connected by a path using only edges with ends in different connected components of the subgraph induced by the subset Y of edges”. Such a formula can be expressed by:

$$\exists U \ U \cap Y = \emptyset \wedge \text{Path}(x, x', U) \wedge \forall u \forall x_1 \forall x_2 [U(u) \wedge \text{endsof}(x_1, x_2, u) \Rightarrow \forall Z \ (\text{CC}_{\bar{Y}}(Z) \Rightarrow \neg(Z(x_1) \wedge Z(x_2)))].$$

Then, condition (a) can be expressed by:

$$\forall X \ [\text{CC}_{\bar{Y}}(X) \Rightarrow (\lceil G \rceil \models \varphi_{\bar{X}})],$$

and condition (b) can be expressed by:

$$\forall X \forall X' (\text{CC}_{\bar{Y}}(X) \wedge \text{CC}_{\bar{Y}}(X') \wedge \text{EDG}(X, X') \Rightarrow \text{Isom}_2(X, X')).$$

and condition (c) by:

$$\forall X \forall x \forall x' [X(x) \wedge X(x') \wedge \text{CC}_{\bar{Y}}(X) \wedge \text{Alternateends}(x, x', Y) \Rightarrow x = x']. \quad \square$$

5. Conclusion

All the proofs in this paper also work for directed graphs. We proved that the composition of an HR set by a finite set of graphs using the cartesian product yields an HR set. We give a concrete construction of a set of “production rules” for the resulting class. Can one obtain an analogous result for VR sets?

The notion of *clique-width* is related to VR grammars (see, e.g., [9]) in the same way that *treewidth* is related to HR grammars. *Treewidth* and *clique-width* are also comparable measures of complexity. Indeed, a property expressible in MS_1 -logic is decidable in linear time on every graph class with bounded *clique-width*. It is interesting to establish upper bounds on the *clique-width* of the product of two graphs expressed in terms of related parameters of the two involved graphs as done in Section 3. Furthermore, other graph operations can be investigated relatively to these aspects, particularly the cartesian sum that we believe is more suitable with VR grammars. Given two graphs G and H , their cartesian sum has vertex set the cartesian product of the two sets V_G and V_H and edge set $\{(x, y), (x', y')\}, [x, x'] \in E_G \text{ or } [y, y'] \in E_H\}$.

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Appendix A. Relationships between tree-decompositions and graph expressions

Graphs and more generally hypergraphs can be obtained as values of terms from well defined (many sorted) *algebras* (see for example, [8,6,1]). Let $\mathcal{C}$ be a countable set of *source labels*. A graph with *sources* is a pair $\langle G, \text{src}_G \rangle$ where G is a graph, $\text{src}_G : \mathcal{C} \rightarrow V_G$ is a total injective mapping and C is a finite subset of $\mathcal{C}$. The set C is called the *type* of G and is denoted by $\tau(G)$. We denote by $\mathcal{G}_{\mathcal{S}}(C)$ the set of all graphs of type C . The set $\mathcal{S}$ of finite subsets of $\mathcal{C}$ is used as the set of sorts. We denote by $\mathcal{G}_{\mathcal{S}}$ the set of all graphs with sources. We consider the following $\mathcal{S}$ -signature:

(1) Parallel composition

If $G \in \mathcal{G}_{\mathcal{S}}(C)$ and $G' \in \mathcal{G}_{\mathcal{S}}(C')$ we let $H = G //_{c, c'} G'$ be the (isomorphism class of the) graph in $\mathcal{G}_{\mathcal{S}}(C \cup C')$ obtained by first taking the union of two disjoint graphs K and K' respectively isomorphic to G and G' and then by fusing any two vertices v and v' that are respectively the c -source of K and the c' -source of K' for some $c \in C \cap C'$.

(2) Forgetting a source label

If $G \in \mathcal{G}_{\mathcal{S}}(C)$ and $c \in C$, we let $H = fg_c(G)$ the graph in $\mathcal{G}_{\mathcal{S}}(C \setminus \{c\})$ with source mapping $src_H = src_G / (c \setminus \{c\})$. In words, the vertex with label c in G becomes “anonymous” in H .

(3) Constant operators

For every source label c , we denote by c the graph with a single vertex which is the c -source, c^ℓ the graph consisting of a single vertex which is the c -source and at which there is a loop and, for every pair $\{c, c'\}$ of distinct source labels, the graph cc' consisting of an edge whose extremities are respectively the c -source and the c' -source.

We let F_{HR} be the $\mathcal{S}$ -signature consisting of $//_{C,C'}, fg_c, c, c^\ell$ and cc' for all relevant C, C' subsets of $\mathcal{C}$ and all source labels c, c' . We obtain an F_{HR} -algebra. The terms in the free algebra $T(F_{HR})$ are called *HR(graph)-expressions*. Every $t \in T(F_{HR})$ denotes a graph $val(t)$ called the value of t . A term t in the free algebra can be viewed as a labeled rooted tree constructed recursively as follows. The root is labeled with the operation f such that $t = f(t_1, \dots, t_{\rho(f)})$, where $\rho(f)$ is the arity of f . For each t_i , we construct a labeled rooted tree in the same way and so on. Hence the leaves are the constants.

For every finite subset C of $\mathcal{C}$, every finite graph G in $\mathcal{G}_{\mathcal{S}}(C)$ is the value of some *HR-expression*. Hence $\mathcal{G}_{\mathcal{S}}$ is homomorphic to the F_{HR} -algebra so defined.

For $\mathcal{K} \subseteq \mathcal{C}$, we denote by $F_{HR}(\mathcal{K})$ the subsignature consisting of the above symbols with $C, C' \subseteq \mathcal{K}, c, c' \in \mathcal{K}$.

Other sets of *HR-operations* have also been considered (see [8,9] for example). It may happen in some proofs, that certain signatures provide more facilities.

The following result has been proved in [7].

$$\text{twd}(G) = \min \{ |C|, C \subseteq \mathcal{C} / G = val(t) \text{ for some } t \in T(F_{HR}(C)) \} - 1.$$

Appendix B. Equational sets of graphs

Equational sets have been defined by Courcelle in the general context of Universal Algebra in a way comparable with context-free languages obtained by concatenation of letters from a fixed alphabet. In order to simplify the readability of this paper, we give the definition in a less general context by considering only what is needed to deal with *HR-equational* sets of graphs. Let F be the F_{HR} signature relative to a countable set of labels $\mathcal{C}$. We define a signature F^+ that allows one to obtain the power-set of $\mathcal{G}_{\mathcal{S}}$ as a (many sorted)-algebra. First, for every $f \in F$, we correspond an $f_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}$ of the same type as follows.

- If f is of type $C \times C' \rightarrow C \cup C'$ then, for any $A_C \subseteq \mathcal{G}_{\mathcal{S}}(C), A_{C'} \subseteq \mathcal{G}_{\mathcal{S}}(C')$, we let $f_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_C, A_{C'}) = \{t //_{C,C'} t', t \in A_C, t' \in A_{C'}\}$.
- If f is of type $C \rightarrow C \setminus \{c\}$ ($c \in C$) then, for any $A_C \subseteq \mathcal{G}_{\mathcal{S}}(C), f_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_C) = \{fg_c(t), t \in A_C\}$.
- For each constant operator f , if t is the term produced by f in $\mathcal{G}_{\mathcal{S}}$ then, the term produced by the corresponding constant operator in $\mathcal{P}(\mathcal{G}_{\mathcal{S}})$ is the singleton $\{t\}$.

We add to the family of operators $(f_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})})_{f \in F}$ two new symbol operators, for every sort $C \subseteq \mathcal{C}$: a symbol $+_C$ of type $C \times C \rightarrow C$ and a constant operator Ω_C of sort C . For every $C \subseteq \mathcal{C}, \Omega_C := \emptyset$, and $A +_C A' := A \cup A'$, for any A, A' subsets of $\mathcal{G}_{\mathcal{S}}(C)$. This ends the definition of the signature F^+ .

Let F denotes again the F_{HR} signature relative to a countable set $\mathcal{C}$ of source labels. A *polynomial system* over F is a sequence of equations $S = \langle u_1 = p_1, \dots, u_n = p_n \rangle$, where $U = \{u_1, \dots, u_n\}$ is an $\mathcal{S}$ -sorted set of variables called the set of the *unknowns* of S . Each term p_i is a *polynomial*, that is a term of the form Ω_C or of the form $t_1 +_C \dots +_C t_m$, where each term t_j is a *monomial* over $F \cup U$ (that is a term in $T(F \cup U)$) of the same sort as u_i .

Let $C_1, \dots, C_n$ be the sorts of $u_1, \dots, u_n$ respectively. A mapping $S_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}$ from $\mathcal{P}(\mathcal{G}_{\mathcal{S}}(C_1)) \times \dots \times \mathcal{P}(\mathcal{G}_{\mathcal{S}}(C_n))$ into itself is associated with S as follows. For any $A_1 \subseteq \mathcal{G}_{\mathcal{S}}(C_1), \dots, A_n \subseteq \mathcal{G}_{\mathcal{S}}(C_n)$,

$$S_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_1, \dots, A_n) := (p_{1\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_1, \dots, A_n), \dots, p_{n\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_1, \dots, A_n)).$$

A *solution* of S in $\mathcal{P}(\mathcal{G}_{\mathcal{S}})$ is an n -tuple $(A_1, \dots, A_n)$, such that for each $i, 1 \leq i \leq n, A_i \subseteq \mathcal{G}_{\mathcal{S}}(C_i)$, and $(A_1, \dots, A_n) = S_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_1, \dots, A_n)$. A solution of S is also called a *fixed-point* of $S_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}$.

For set-inclusion, the least solution (in $\mathcal{P}(\mathcal{G}_{\mathcal{S}})$) of such a system S is denoted by $(L((S, \mathcal{G}_{\mathcal{S}}), u_1), \dots, L((S, \mathcal{G}_{\mathcal{S}}), u_n))$. An *HR-equational* set of graphs is a component of such a least solution. It is obtained as an infinite set-union:

$$L((S, \mathcal{G}_{\mathcal{S}}), u_i) = \bigcup_{l \geq 0} A_i^l,$$

where $A_i^0 = \emptyset$, and $(A_1^{l+1}, \dots, A_n^{l+1}) = S_{\mathcal{P}(\mathcal{G}_{\mathcal{S}})}(A_1^l, \dots, A_n^l)$.

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